

Muhammad Ahmed Khan

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[GitHub Profile](#)

PROFESSIONAL SUMMARY

Computer Science student (2024–2028) with hands-on experience in software development, database management, workflow automation, and data analysis. Proficient in Java, Python, and SQL with practical exposure to OOP, DSA, and cloud-based systems. Seeking opportunities to apply technical skills in real-world environments.

TECHNICAL SKILLS

Programming Languages: Java, Python, SQL

Frameworks & Libraries: JavaFX, Swing, Pandas, Matplotlib

Databases & Cloud: Azure SQL, SQL Server (SSMS), File-based storage

Tools & Platforms: n8n (Workflow Automation), Microsoft Excel, Git, GitHub

Concepts: Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), Database Normalization (1NF–3NF), CRUD Operations, ERD Design, Sorting Algorithms, Memory Allocation

Soft Skills: Communication, Team Collaboration, Problem Solving

AI Tools & Platforms: Claude (Anthropic), Gemini (Google), ChatGPT (OpenAI), Antigravity, GitHub Copilot

EDUCATION

Bachelor of Computer Science 2024 – 2028 (Expected)

Karachi School of Business and Leadership (KSBL)

EXPERIENCE

Campus Ambassador – *atomcamp* – KSBL – Present

- Promoted Atomcamp's data science and AI programs to the student community at KSBL.
- Coordinated outreach activities, workshops, and tech events to drive student engagement.
- Served as the primary point of contact between Atomcamp and KSBL students.

PROJECTS

FasalAI – AI-Powered Agricultural Export Compliance Platform

Multi-agent AI system that guides Pakistani farmers and exporters through SPS, MRL, and phytosanitary compliance for international markets (EU, USDA, Gulf). Built for the National AI Hackathon powered by Atomcamp.

- **Multi-agent architecture using CrewAI:** Designed and built 5 specialist agents — MRL Checker, SPS Standards, Document Generator, Risk Flagger, and Urdu Language Agent — each with dedicated tools and clearly defined tasks.
- **RAG pipeline using ChromaDB and LangChain:** Embedded and queried EU Regulation EC 396/2005, IPPC phytosanitary standards, Pakistan DPP rules, and Gulf GCC food safety documents — enabling sub-2-second compliance lookups.
- **Live API integration:** Connected EU Pesticides DB, RASFF border alert API, and USDA FAS MRL database to cross-check pesticide residue limits across 50+ export markets in real time.
- **Urdu language agent:** Translates complex regulatory output into plain Urdu, making the system accessible to smallholder farmers with limited English literacy.
- Generated pre-filled export documents (phytosanitary certificates, certificates of origin) using IPPC ePhyto Hub templates, reducing document preparation time from days to minutes.
- Delivered a working end-to-end demo: farmer inputs crop + destination + pesticides used → system detects violations, suggests compliant alternatives, and outputs a full compliance report in Urdu.

n8n Workflows

- **Marketing Multi-Agent System** — Built a multi-agent marketing system with three specialist agents (Brand, Content, Paid Ads) running in parallel branches off a shared campaign brief. Used Merge nodes to synchronize agent outputs and generate a consolidated executive summary, avoiding brittle code-based transformations.

- **Conversion Rate Optimizer** — Developed a conversion rate optimization workflow that scrapes a landing page, runs a 3-agent chain (idea generation, parameter setting, and judging) to produce and score improvement ideas, and implements a feedback loop that automatically retries with the critic's suggestions when scores fall below a quality threshold.
- **Lead Generation Pipeline** — Created a lead generation workflow integrating the Apify API to scrape Google Maps for business listings, then looped through business websites in parallel data streams to extract contact emails, merging multiple branches into a single deduplicated output.
- **Automation of Multi-Year Excel Data Processing** — Built an automated n8n workflow to filter supplier-specific data from multi-year, month-wise Excel files, reducing manual processing time by 3–4 hours by consolidating repetitive filtering into a single automated workflow. Handled multiple folders and file formats (.xls, .xlsx), improving data processing efficiency and accuracy.

College Admission System – Java (OOP)

- Developed a role-based college admission system using Java OOP principles and Swing/JavaFX UI.
- Implemented modules for university applications, scholarship management, and entry test processing.
- Applied OOP concepts including inheritance, polymorphism, abstraction, and interfaces.
- Integrated file handling and database storage for persistent data management.

Sorting and Memory Allocation Simulator (DSA)

- Built a JavaFX GUI to visually simulate sorting algorithms: Bubble, Selection, Insertion, and Merge Sort.
- Developed a memory allocation simulator with Best Fit, First Fit, and Worst Fit strategies and live fragmentation visualization.
- Implemented user controls (start, pause, reset, randomize) with optimized performance for large input sizes.

Movie Cinema Management System (DBMS)

- Migrated a file-based cinema system to a relational SQL database hosted on Microsoft Azure.
- Designed an Entity-Relationship Diagram (ERD) and normalized tables to 3NF for Movies, Shows, Seats, and Bookings.
- Implemented CRUD operations, stored procedures, and triggers for automated booking and seat management.
- Connected Azure SQL database to a front-end application using SQL Server Management Studio (SSMS).

CERTIFICATIONS

- OOP – University of California, San Diego
- Data Structures and Algorithms (DSA) – University of California, San Diego
- Building APIs in Python – [Datacamp](#)
- Introduction to AI Agents - [Datacamp](#)
- Introduction to Workflow Automation with n8n - Datacamp
- Intermediate workflow Automation with n8n - [Datacamp](#)
- Building marketing workflows with n8n - [Datacamp](#)
- Claude Code 101 - [Datacamp](#)
- Microsoft Excel – NS Training
- Communication & Soft Skills – Digiskills
- Introduction to Claude Models – Anthropic (In Progress)

EXTRACURRICULAR ACTIVITIES

- Participated in GDG Kolachi Developers Day
- Attended Power BI Masterclass by AI Datayard
- Participated in devDay'26 - SQL showdown
- Participated in National AI hackathon